GROUP WORK PROJECT #1
GROUP NUMBER: 5805

MScFE 560: FINANCIAL MARKETS

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Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed. **Note:** You may be required to provide proof of your outreach to non-contributing members upon request.

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Task 1:

1) Income stocks (MD RASHEDUL ISLAM RASEL)

Portfolio Fundamental:

- In order to gain exposure to various growth opportunities and market risks, portfolio A was diversified across three sectors: Healthcare, Communication Services, and Information Technology. The stocks are relatively Cigna Corporation (CI), Netflix Inc. (NFLX) and NVIDIA Corporation (NVDA).
- Data Frequency: Daily
- Trading Days: 252 days per year.

Statistical Overview:

- **Average Return:** NVDA has a higher return than CI and NFLX, which is 45.51%.
- **Standard Deviation:** NFLX and NVDA are highly volatile stocks, indicating their risk level is higher than CI.
- **Skewness:** CI and NVDA exhibit close to zero skewness. NFLX display significantly negative skewness, suggesting the return is skewed left.
- **Kurtosis:** NFLX has the highest kurtosis, suggesting higher returns than other stocks of the portfolio.

	CI	NFLX	NVDA
Average	0.242333	0.103877	0.455077
Standard Deviation	0.374883	0.518577	0.559768
Skewness	-0.077398	-1.639958	0.038876
Kurtosis	9.067549	21.426913	2.110989

Correlation Matrix:

The term "correlation" refers to the measuring of a connection. It shows how asset classes have historically been linked to each other. We understand that each item in our portfolio has some degree of interdependence. They are positively associated if they travel in the same direction and negatively correlated if they move in the opposite direction. We are one step closer to optimising our earnings and/or decreasing our risk by interpreting the correlation matrix.

	CI	NFLX	NVDA
CI	1	0.157538	0.350285
NFLX	0.157538	1	0.522988
NVDA	0.350285	0.522988	1

Covariance Matrix:

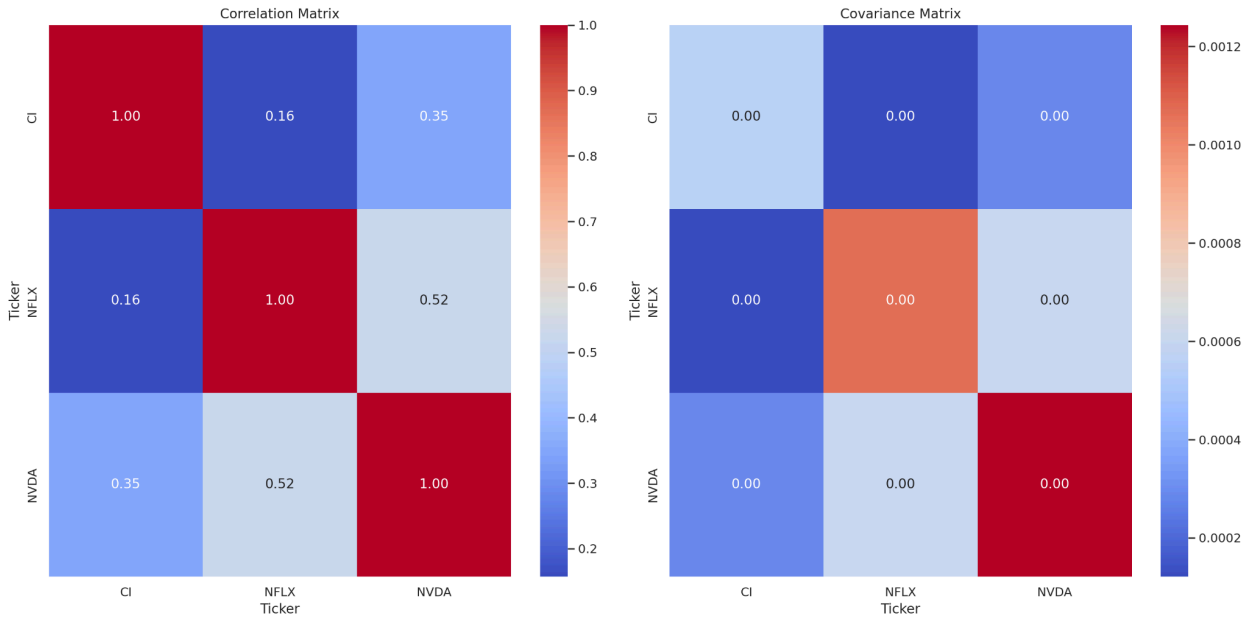
Covariance measures how two variables vary together. A positive covariance means they move in the same direction, and a negative covariance means the opposite. We can see the portfolio's stocks are moving in the same direction. The magnitude of covariance shows the strength of the relationship between the variables.

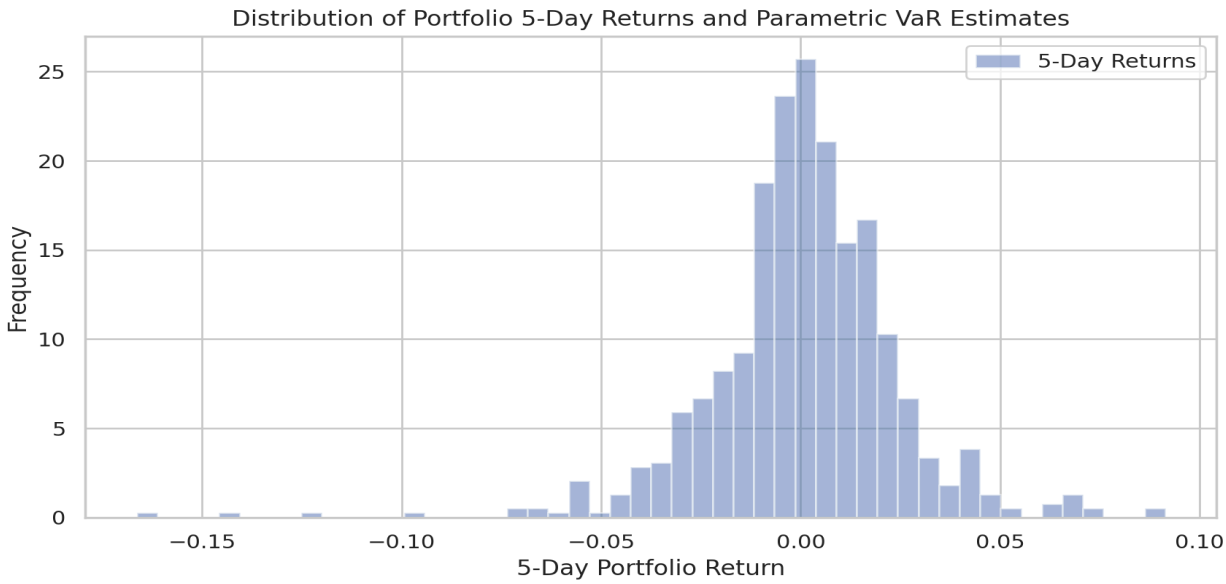
	CI	NFLX	NVDA
CI	0.000558	0.000122	0.000292
NFLX	0.000122	0.001067	0.000602
NVDA	0.000292	0.000602	0.001243

NVDA Stock Price and Dividends
Price
Date
300
250
200
150
100
50
2020-01
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01

CI Stock Price and Dividends
Price
Date
300
250
200
150
2020-01
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01

NFLX Stock Price and Dividends
Price
700
600
500
400
300
200
2020-01
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01


Correlation Matrix
CI
NFLX
NVDA
Ticker
1.00
0.16
0.35
0.16
1.00
0.52
0.35
0.52
1.00
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
Covariance Matrix
0.00
0.00
0.00
0.00
0.00
0.00
0.00
0.00
0.00
0.0012
0.0010
0.0008
0.0006
0.0004
0.0002


Distribution of Portfolio 5-Day Returns and Parametric VaR Estimates
5-Day Returns
Frequency
5-Day Portfolio Return
0
5
10
15
20
25
−0.15
−0.10
−0.05
0.00
0.05
0.10

Portfolio Construction:

The portfolio value was initially fixed at $1,000,000. As we previously examined, we created a portfolio for the three assets CI, NFLX, and NVDA (assumed to have equal weights of around 0.3333). Along with an annualised covariance matrix to evaluate overall risk and inter-asset correlations, a 5-day rolling return is computed for short-term performance evaluation using daily log returns. Value at Risk (VaR) and the portfolio standard deviation are computed to detect possible dangers. VaR estimates at 90%, 95%, and 99% confidence levels forecast **maximum losses of up to $122,965.35** under unfavourable circumstances. By using the diversification advantages of the chosen equities, this method promotes prudent asset allocation and improves risk management.

Confidence Level	Value at Risk
90%:	**$ 67,739.84**
95%:	**$ 86,943.15**
99%:	**$122,965.35**

Portfolio Analysis: Short Selling and Credit Risk:

- **Short Selling:** As the assets in this portfolio are CI, NFLX, and NVDA—**can be sold short**. If we talk about its mechanics we have to follow a few steps to short the stocks. First, borrow these stocks from a lender. The borrower predicts these stock prices will go down at some point in the future and has to buy that trade a cover. On the other hand, if the price of these stocks goes up, the short seller has to buy at a high price, and he/she loses money because sold low and bought high price. Lastly, borrowers have to return these stocks to the lender. As we can see, NVDA and CI pay dividends to stockholders. So, the short seller also has to pay dividends. During the time in which they are borrowing the stock, they have to pay a small amount known as a financial fee.
- **Credit Risk:** The portfolio of CI, NFLX, and NVDA stocks we have chosen does not have credit risks like fixed-income securities or loans. However, these stocks carry other risks, such as market, business, and systemic risks.

2) Sovereign and corporate bonds (Xing Xin)

Portfolio Fundamental:

Portfolio B has five primary assets:

- United States 10 Year Government Bonds Yield (US10Y)
- United States 2 Year Government Bonds Yield (US02Y)
- Germany 10 Year Government Bonds Yield (DE10Y)
- Japan 10 Year Government Bonds Yield (JP10Y)
- Bank of America CDS 5 Year USD (BAC5YUSAX=R)

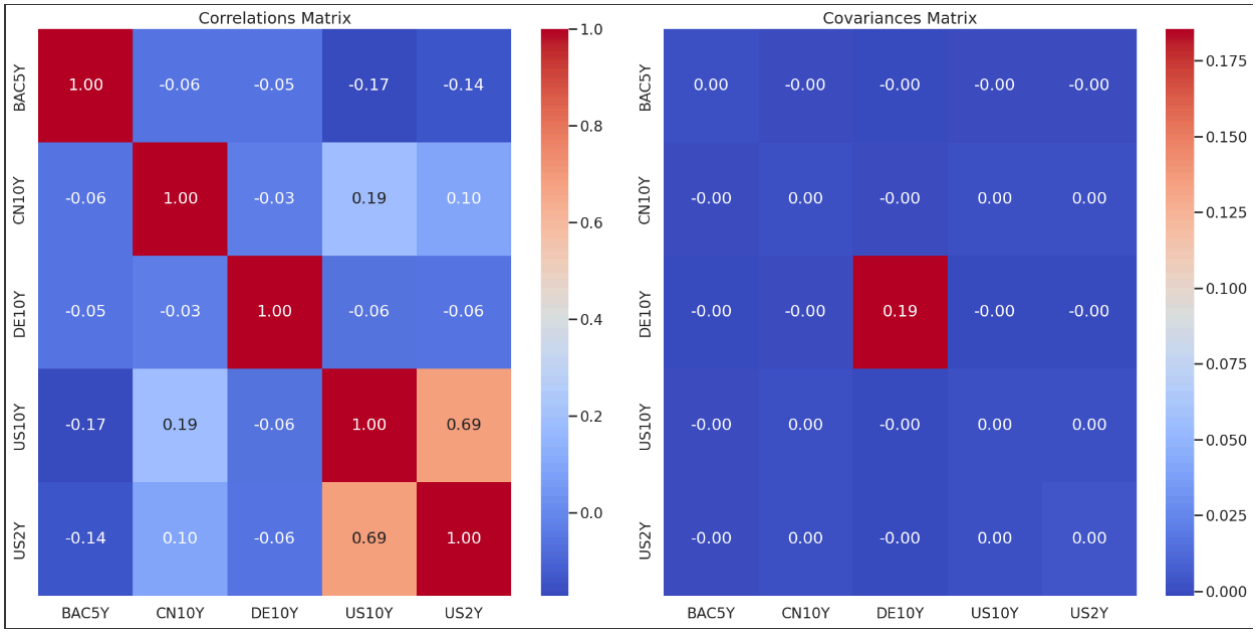
Data Frequency: Daily.

Trading Days: 252 days one year.

Statistical Overview:

Bonds	Annual Return	Volatility	Skewness	Kurtosis
BAC5YUSAX	0.441123	0.613063	1.293829	9.244892
CN10Y	0.441123	0.120128	-0.696920	8.664481
DE10Y	-1.458534	6.835668	-13.653749	291.464276
US10Y	0.476142	0.725804	1.425941	23.709645
US02Y	0.757846	0.949877	0.653340	7.897411

Correlation Matrix & Covariance Matrix:



Portfolio Construction:

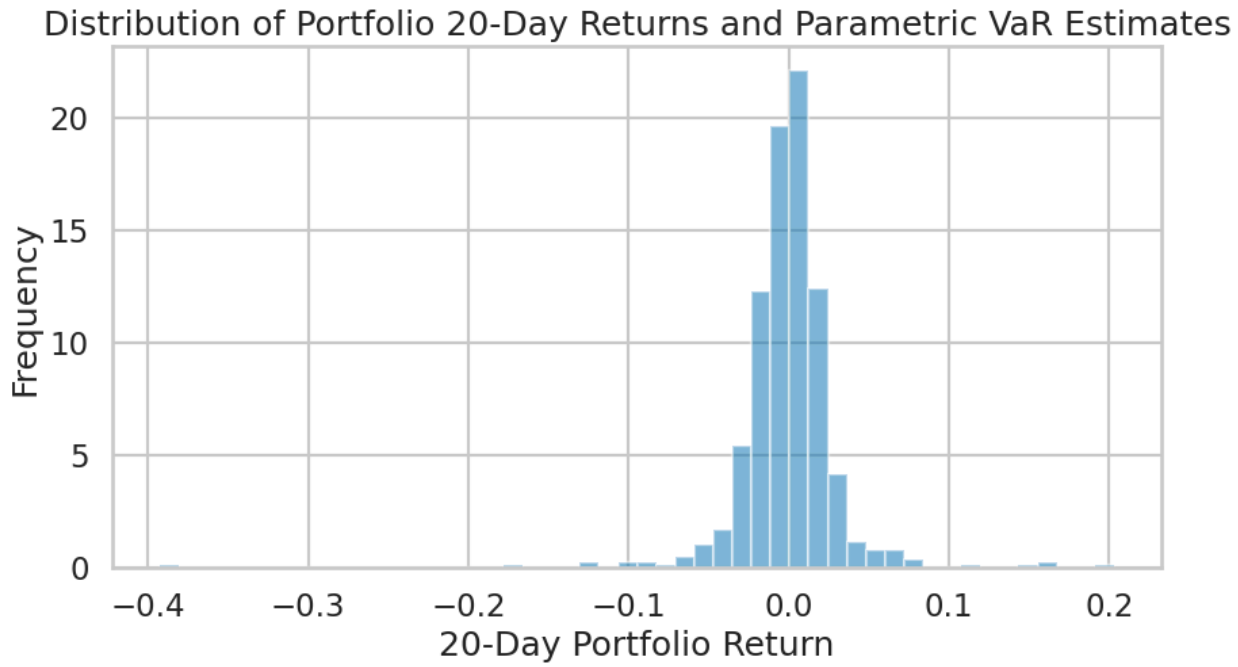
The initial capital of this portfolio is set at $1,000,000.

This portfolio contains 5 assets, with the weight of each asset set to:

[BAC5Y: 0.2, CN10Y: 0.2, DE10Y: 0.2, US10Y: 0.1, US02Y: 0.1]

We use a month's (20 trading days) return to measure the performance of this portfolio. We use standard deviation and value at risk (VaR) to calculate and identify potential risks, with VaR assessed at 90%, 95%, and 99% confidence levels, predicting maximum losses of up to $332,962.57 in adverse conditions.

Confidence Level	Value at Risk
90%:	$183,424.29
95%:	$235,422.53
99%:	$332,962.57



Portfolio Analysis: Short Selling and Credit Risk:

Shorting

1. Can this portfolio be sold short?

   Yes.

2. If it can be sold short, explain the mechanics.

   There are two main ways to sell short:

   a. **Direct Short Selling**: This involves borrowing individual bonds from a holder and selling them in the market. You'd then need to buy them back later to return them to the lender, profiting if the price falls in the meantime. This method requires a margin account and can be complex due to factors like interest payments on the borrowed bonds.

   b. **Derivatives**: Financial instruments like futures contracts and inverse bond ETFs offer indirect ways to short a bond portfolio. These allow you to speculate on price movements without directly owning the underlying bonds. Inverse ETFs, for instance, are designed to return the opposite of a specific bond index, so you profit if the index goes down.

Credit Risk

1. Does this portfolio have credit risk?

   Yes, this portfolio has credit risk but the risk level is very low, because the five assets have three issued from governments and another one is issued by the Bank of America. The government and Bank of America are assumed to have the ability to raise tax revenue to repay their debts, but there are a few scenarios where the government and bank might carry some credit risk.

2. If it has credit risk, explain what the credit risk is for your portfolio.

   a. **Bonds Issued by Foreign Governments:** While governments of developed nations are considered reliable borrowers, there's a slightly higher chance of default on bonds issued by governments in developing economies or those with unstable political situations.

   b. **Eventualities with Extreme Economic Stress:** In an extremely unlikely scenario, a government could face such a severe economic crisis that it might be unable to meet its debt obligations. This is a remote possibility for most developed countries, but it can't be entirely ruled out.

3) Cryptocurrencies (Le Nguyen Dang Khoa)

Portfolio Fundamental:

- Portfolio C has three primary assets: BTC (Bitcoin), ETH (Ethereum), and XRP (Ripple).I selected for their significant market capitalizations[8] and the transparency and integrity of their data, which minimizes risks related to manipulation and bias.
- Data Frequency: Daily data is used, from lectures that cryptocurrencies trade 24/7[11].
- Trading Days: The analysis assumes 365 trading days per year.

Statistical Overview:

- Average Returns: BTC (0.383594), ETH (0.859648), and XRP (0.633218) show varied performance levels, with **ETH yielding the highest average returns**.
- Standard Deviation: Indicates risk levels, with **XRP (1.017885) having the highest volatility**, followed by ETH (0.824959) and BTC (0.609552) . This is reasonable respect that shows the respect of market capitalizations .
- Skewness: BTC and ETH display negative skewness, suggesting returns are skewed left. Conversely, XRP exhibits positive skewness, indicating potential for unusually high returns.
- Kurtosis: **All currencies show high kurtosis**, especially XRP (15.161942), indicating a higher likelihood of extreme values.

Correlation Matrix:

- **BTC and ETH have a high correlation (0.817242)**, suggesting similar price movements.
- XRP's correlation with BTC (0.586553) and ETH (0.609132) is lower, which could be advantageous for diversification.

Covariance Matrix:

- Reflects the covariance among the assets, with the highest between ETH and XRP (0.002030), indicating their returns move more similarly compared to others.

Figure 1.3.1: Crypto Prices Comparison

	BTC	ETH	XRP
Average	0.383594	0.859648	0.633218
Standard Deviation	0.609552	0.824959	1.017885
Skewness	-0.757178	-0.373377	1.327285
Kurtosis	10.878126	8.027659	15.161942

Table 1.3.1: Statistical Overview

	BTC	ETH	XRP
BTC	1	0.817242	0.586553
ETH	0.817242	1	0.609132
XRP	0.586553	0.609132	1

Table 1.3.2: Correlation Matrix

	BTC	ETH	XRP
BTC	0.001474	0.001631	0.001444
ETH	0.001631	0.002701	0.002030
XRP	0.001444	0.002030	0.004111

Table 1.3.3: Covariance Matrix

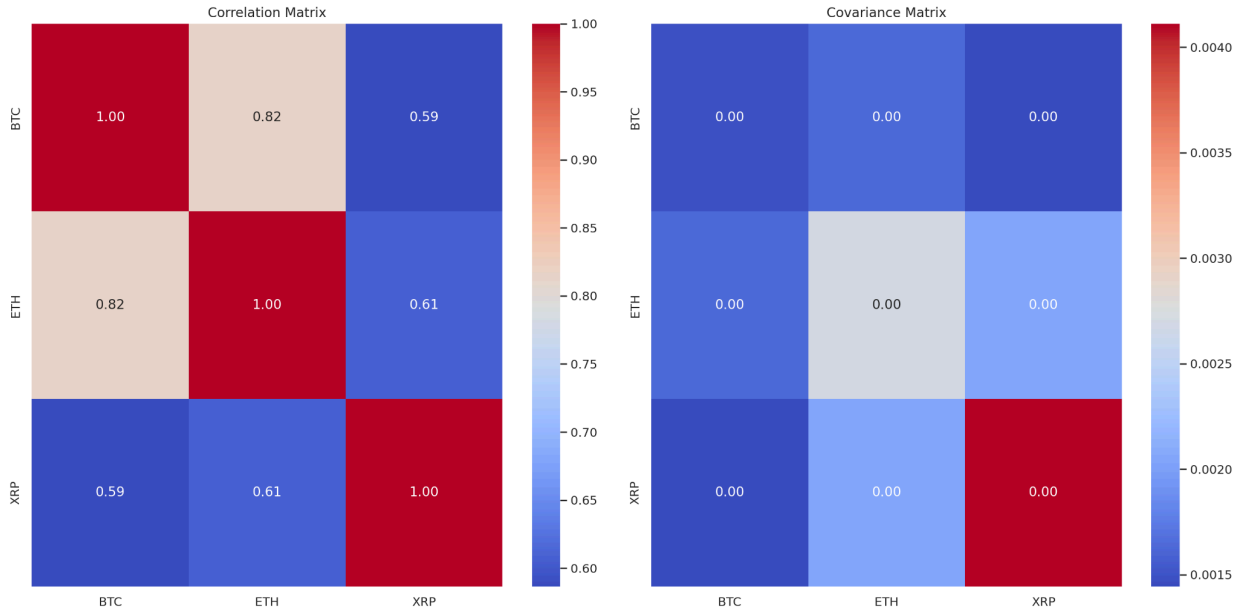


Figure 1.3.2: Correlation and Covariance heatmap

Portfolio Construction:

Initially set at $1,000,000 for the portfolio value. I design a portfolio for 3 assets as we analyzed before: BTC, ETH, and XRP (given the equal weights of approximately 0.3333 are assigned to each asset). Daily log returns are utilized to compute a 5-day rolling return for short-term performance evaluation, alongside an annualized covariance matrix to assess inter-asset relationships and overall risk. Portfolio standard deviation and Value at Risk (VaR)[1][2] are calculated to identify potential risks, with VaR assessments at 90%, 95%, and 99% confidence levels projecting **maximum losses up to $240,816.26** in adverse

conditions. This strategy enhances risk management and supports informed asset allocation by capitalizing on the diversification benefits of the selected cryptocurrencies.

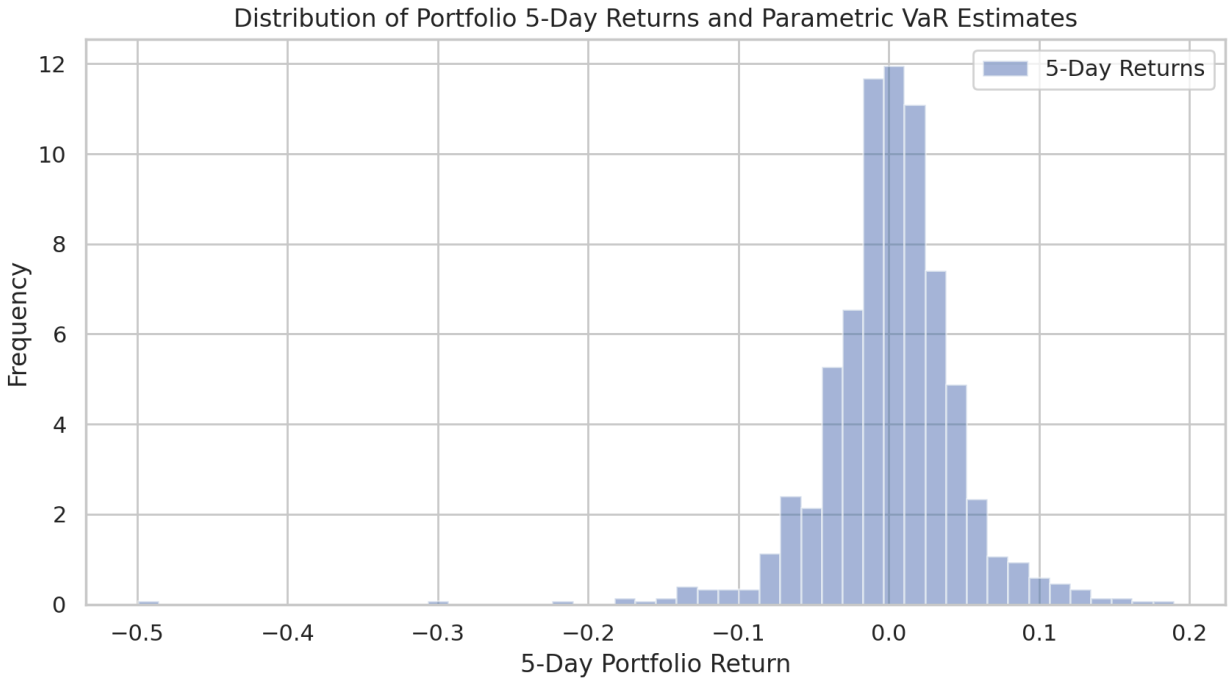


Figure 1.3.3: 5-Day Portfolio Return

Portfolio Analysis: Short Selling and Credit Risk:

- **Short Selling:** As the assets in this portfolio are BTC, ETH, and XRP—**can be sold short**. The mechanics of short selling in cryptocurrencies involves borrowing the digital assets from an exchange(Binance, Bitget, Bybit,...) and immediately selling them at the current market price (All these actions are in futures account). The goal is to earn profit from the difference between the price at time we place order and the time we close order. However, this strategy involves significant risk, particularly due to the high volatility of cryptocurrencies, which can lead to substantial losses if the market moves against the position, and your account does not have sufficient balance to maintain if the price goes too high.
- **Credit Risk:** Inherent credit risk exists within this cryptocurrency portfolio, although it differs from the traditional credit risk associated with bonds or loans. Credit risk primarily arises from the possibility of the counterparty, such as the exchange or wallet provider, failing to fulfill their obligations. This failure could result from operational issues, bankruptcy, fraud, or regulatory changes. For instance, if an exchange holding the assets goes bankrupt or is hacked (similar to the FTX situation), investors may lose their investment without any recourse.

Task 2:

1. **Our Group distribute the following roles among our team members:**
   - Student A will play the technical interviewee: Xing Xin

- Student B will play the non-technical interviewee: Khoa
- Student C will play the role of journalist: Rashedul

2. Interview section

Rashedul: Hello, Xing Xin and Khoa. Today, we're exploring different aspects of asset management and how specific financial regulations impact the finance sector. Let's start with some specifics.

Rashedul: Khoa, could you elaborate on the liquidity risks associated with income stocks in your portfolio compared to other asset classes?

Khoa: Yes, in my point of view, income stocks often have liquidity risk compared with crypto assets or even the large-cap stocks. For example, stock trading days are just 252 days a year compared with crypto assets, which can trade 24/7(Even crypto has much more market capital). Moreover, I consider that trading stock is often limited in that country only because we hardly have access to the exchange of other countries. In addition, income stock trade less active compare with large one, if you invest a huge amount on an industry it would easier for you to sell a bunch of stocks large-cap because it have more liquidity.

Rashedul: Interesting. Xing, on a related note, what criteria do you use to select these income stocks for diversification and stability in your strategy?

Xing Xin:

For the diversification and and stability, I usually use these criteria:
Diversification:

- **Geographic Location**: Invest in companies from developed and emerging markets to hedge against currency fluctuations and economic conditions specific to a region.
- **Industry Sector**: Spread my investments across different industries. This way, a downturn in one sector won't cripple my entire portfolio. Look for industries with different growth cycles (e.g., consumer staples vs. technology).
- **Company Size**: Include a mix of large-cap (established, stable companies), mid-cap (potential for growth), and small-cap stocks (higher risk, higher reward) depending on my risk tolerance.

Stability:

- **Financial Strength**: Look for companies with a solid track record of profitability, low debt levels, and healthy cash flow. This indicates their ability to weather economic downturns.
- **Dividend History**: Companies with a history of paying consistent dividends provide a steady stream of income and tend to be more mature and stable.
- **Beta**: Beta measures a stock's volatility compared to the overall market. A beta below 1 indicates the stock is less volatile than the market, making it a good choice for stability.

Rashedul: Thanks, both of you. It's insightful to see how technical and strategic considerations intertwine. Moving to bonds, Khoa, can you provide more details on how you manage the leverage of sovereign and corporate bonds in your portfolio?

Khoa: To be honest I'm a risk averse so I prefer to invest in risky assets like stocks or crypto. issuer's stability and the quality of fund management. If I were to consider bond investments, I would carefully assess the risk-reward ratio and ensure alignment with my risk tolerance. This involves evaluating factors such as credit ratings, economic conditions, and overall market stability before making any investment decisions.

Rashedul: Xing, please provide more information about how you manage the leverage of sovereign and corporate bonds in your portfolio. I'm curious to know how this affects their valuation and performance.

Xing Xin: Leverage refers to the amount of debt a government or a company issues compared to its equity. The higher the leverage, the greater the risk of default but also the potential for higher returns. To managing the leverage I usually consider:

- **Credit Ratings:** Sovereign and corporate bonds are assigned credit ratings by agencies like S&P. These ratings reflect the issuer's creditworthiness and ability to repay debt. Lower-rated (higher risk) bonds typically offer higher yields to compensate investors for the increased risk.
- **Portfolio Allocation:** Balance my portfolio with a mix of investment-grade bonds (higher credit ratings, lower risk, lower yields) and high-yield bonds (lower credit ratings, higher risk, higher yields) based on my risk tolerance and income needs.
- **Maturity Ladder:** Stagger my bond maturities across different time horizons. This helps me manage interest rate risk and provides a predictable stream of income as bonds mature.

For the impact on valuation and performance, I think bond prices and interest rates have an inverse relationship. When interest rates rise, existing bonds with lower fixed coupons become less attractive, driving their prices down. Higher leverage can amplify these price movements for leveraged issuers' bonds.

Rashedul: Leverage and selection seem to play significant roles in bond management. It's fascinating to see how each of you approaches these aspects. Talking about cryptocurrencies, how do they vary from traditional financial products, and what possible risks are involved in trading or investing in them?

Xing Xin: You're absolutely right. Leverage and selection are fundamental aspects of effective bond management. It's interesting to see different approaches, and likewise, the world of cryptocurrencies offers unique considerations compared to traditional financial products. Here's a breakdown of the key differences and potential risks involved:

- **Underlying Asset:** Traditional financial products like stocks and bonds represent ownership or debt in a real-world asset (company, government). Cryptocurrencies don't have a physical backing and derive their value purely from market demand and confidence.
- **Regulation:** Traditional financial markets are heavily regulated, with oversight bodies ensuring transparency and investor protection. Cryptocurrency markets are largely unregulated, leading to higher volatility and potential for scams.
- **Transaction Processing:** Traditional financial transactions are processed by central institutions. Cryptocurrencies rely on decentralized networks (blockchain) for verification and record-keeping, which can be slower and more complex.

The risk of trading or investing in Cryptocurrencies are diversity, here are some significant risks:

- **Potential for Illegality**: Cryptocurrencies have been used for illegal activities due to their pseudonymous nature. Regulatory crackdowns could impact their value and usage
- **Uncertain Future Regulation**: Governments are still grappling with how to regulate cryptocurrencies. Future regulations could significantly impact their value and use cases.
- **Volatility**: Cryptocurrencies are notoriously volatile, experiencing significant price swings in short periods. This can lead to substantial losses for investors.
- **Security**: Cryptocurrency exchanges and wallets can be vulnerable to hacking, resulting in theft of your holdings.

Rashedul: And Khoa, from your perspective, what non-technical factors do you consider when assessing the potential of cryptocurrencies?

Khoa: Beside the technical indicators i often use when place an order i often have to consider many other aspects:

Is the team behind the token good, does the project have valuable insights and realistics, …

Rashedul: Cryptocurrencies certainly add a unique dimension to asset management, with both technical and broader market considerations impacting decisions. Now, moving onto regulations. Xing Xin, can you outline the specifics of the new regulation we are discussing today?

Xing Xin:

Sure, I think the specifics of the new regulation we discussed today include:

No single, new regulation: Proposals like the Financial Innovation and Technology (FIT) for the 21st Century Act haven't been passed yet.

Focus on existing agencies: The Securities and Exchange Commission (SEC) and the Commodity Futures Trading Commission (CFTC) are the main regulatory bodies for now.

Proposed SEC rules: The SEC has proposed some new rules:

Investment advisors may need to hold client crypto assets with qualified custodians, potentially excluding current crypto exchanges.

The SEC might expand the definition of regulated exchanges to include crypto entities.

Rashedul: And Khoa, considering this regulation, how wide is its scope? Is it intended just for national levels, or does it reach globally?

Khoa: The crypto regulation varies between countries. For my country Vietnam, the government has not protected the crypto investors[9]. This lack of protection for crypto investments mirrors situations seen in forex trading, where regulatory frameworks may lag behind technological advancements, leading to uncertainty and potential risks for investors. But in the big picture crypto was widely accepted by many countries and should be considered a legal investment like the case of Bitcoin ETFs [10], El Salvador and the Central African Republic do accept crypto [5], ...

Rashedul: Discussing the potential pitfalls, Xing Xin, what are some of the downsides or unintended consequences that might arise from this regulation?

Xing Xin: [Describes possible downsides]

Market Manipulation:

The crypto market is susceptible to manipulation, including stablecoins. Large transactions or coordinated efforts can impact stablecoin prices.

Regulatory authorities are concerned about market integrity and may implement measures to prevent manipulation.

Systemic Risk:

If a widely used stablecoin were to fail or experience a sudden loss of value, it could have broader implications for the financial system.

Regulatory oversight aims to mitigate systemic risks associated with stablecoins.

Rashedul: Khoa, could you compare these downsides to the intended effects Xing Xin mentioned? What stands out?

Khoa: Absolutely, Rashedul. From my perspective, the downsides of the new regulatory frameworks for cryptocurrencies, while aimed at increasing transparency and security, raise concerns about the potential dilution of the fundamental principles of **decentralization[12]** and **resistance to manipulation**. This could lead to a decrease in investor interest, as the inherent

appeal of cryptocurrencies lies in their ability to operate independently of traditional financial systems.

Rashedul: Lastly, regulatory changes are a constant in financial markets. Xing, how do these changes affect your portfolio's asset classes, and how do you manage liquidity and mitigate risks?

Xing Xin: New regulations can directly impact specific asset classes. For example, stricter capital requirements for banks could reduce their participation in certain markets, impacting liquidity and potentially raising borrowing costs for corporations (affecting corporate bonds).Regulations can also have indirect effects. For instance, environmental regulations might make certain industries less attractive, impacting the value of stocks in those sectors.

I have some strategies investors to manage portfolio risk and liquidity in the face of regulatory changes:

- **Diversification**: Spread your investments across different asset classes to mitigate the impact of regulatory changes on any single asset class.
- **Stay Informed**: Keep yourself updated on regulatory developments that could affect your portfolio. Financial news outlets and industry publications are good sources of information.
- **Liquidity Buffer**: Maintain a portion of your portfolio in highly liquid assets (cash, short-term bonds) to meet unexpected needs or capitalize on new opportunities arising from regulatory shifts.

Rashedul: And Khoa, how do you manage risks and investments in such a dynamic regulatory environment?

Khoa: In such a dynamic regulatory environment, managing risks effectively requires a multifaceted approach.

Firstly, I stay continuously informed about new regulations and their implications through reliable financial news sources, such as CoinDesk, and consultations with legal experts. This helps in anticipating changes and adjusting strategies promptly.

Additionally, diversification plays a crucial role in my investment strategy. By spreading investments across different asset classes, including cryptocurrencies(large market cap: BTC ETH, XRP,...), stocks(large-cap, ETFs, …), the impact of adverse regulatory changes on any single investment is mitigated. Furthermore, I utilize advanced risk management tools and techniques such as stop-loss orders when I trade futures and position sizing(always set 1/20 capital) to manage the volatility associated with crypto investments.

Rashedul: Khoa, what were the key events of the FTX crisis that led to increased regulatory scrutiny in the cryptocurrency market?

Khoa: The FTX crisis unfolded rapidly, starting when concerns were raised publicly about the exchange's financial health. These concerns were initially triggered by a tweet from a rival exchange's CEO, which cast doubt on FTX's liquidity[13]. This sparked a widespread panic among FTX customers, leading to a massive withdrawal spree. Unfortunately, FTX was unable to meet these withdrawal demands, confirming fears of its financial instability and mismanagement.

Rashedul: Thank you both for your insights today. Your detailed responses provide a deep understanding of how regulations, market dynamics, and strategic management intertwine in the field of asset management.

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